



Darstellung des Themas und Skizzierung des Forschungsvorhabens

Description of the topic and outline of the research project

Anlage zum Antrag auf Zulassung zum Promotionsverfahren in der MIN-Fakultät

Attachment to the application for admission to doctoral procedures at the MIN Faculty

A. Antrag | *Application*

Name, Vorname der Antragstellerin bzw. des Antragstellers | *Surname, first name of applicant:*

Damerow, Hauke

Vorläufiger Arbeitstitel (in Deutsch oder Englisch) | *Working title (in German or English):*

Novel Ab Initio Methodology for Trions and Biexcitons in Moiré and Janus 2D materials

Ort
Place

Datum
Date

Unterschrift der Antragstellerin bzw. des Antragstellers
Signature of applicant

B. Zustimmung der Betreuenden / der Betreuungskommission | *Supervisor's or supervision panel's consent*

Ich / wir befürworten die vorgelegte Skizzierung des Promotionsvorhabens.

I / we approve the outlined dissertation project.

Betreuerin bzw. Betreuer | *Supervisor*

Datum
Date

Name und Unterschrift der Betreuerin bzw. des Betreuers
Name and signature of supervisor

Co-Betreuerin bzw. Co-Betreuer | *Co-supervisor*

(falls zutreffend) | *(if applicable)*

Datum
Date

Name und Unterschrift der Co-Betreuerin bzw. des Co-Betreuers
Name and signature of co-supervisor

Vorsitzende bzw. Vorsitzender | *Chair*

(falls zutreffend) | *(if applicable)*

Datum
Date

Name und Unterschrift der Vorsitzenden bzw. des Vorsitzenden der Betreuungskommission bzw. einer weiteren Co-Betreuerin bzw. eines weiteren Co-Betreuers
Name and signature of chair of the supervision panel or a further co-supervisor



C. Zusammenfassung zum Thema und zum Forschungsvorhaben | *Outline of the topic and research project*

Die Zusammenfassung kann in Deutsch oder Englisch erstellt werden und soll maximal zwei Seiten (Arial 12pt) umfassen. Text bitte in das untere Feld eintragen oder an dieses PDF-Formular anhängen. Sofern Letzteres, dann das vorliegende Formular (vollständig unterschrieben) und die PDF-Datei mit der Zusammenfassung zu einer PDF-Datei zusammenfügen.

Please note that the outline can be written in German or English and should not exceed two pages (Arial, 12pt). Please enter outline in the field below or attach to this PDF form. If the latter, then merge the present form (fully signed) and the PDF file with the outline into one PDF file.

Low-dimensional 2D materials have attracted a huge interest from the international research community in recent years. They are considered a key class of materials for studying fundamental quantum mechanical effects, e.g. unconventional superconductivity. They are also a key material class for many potential future applications, ranging from more effective solar cells via faster computation, energy storage, up to quantum computing. 2D materials can be created by exfoliation from the bulk material; characteristic of them are the strong covalent bonds in the 2-dimensional layers and weaker van der Waals forces between the different layers.

The physical properties of 2D materials and their heterostructures differ strongly from those of their bulk material form.

Transition metal dichalcogenides (TMDs) form a subclass of 2D materials. They are considered very versatile, as they can have a direct band gap in the visible wavelength range and show strong spin-orbit coupling.

Since the formation of twisted 2D bilayer graphene, which is superconducting only at different 'magic' twist angles between the stacked and one slightly rotated graphene monolayers, there has been a growing research activity on these so-called Moiré materials. The twist angle is a new degree of freedom which allows significant changes in the electronic band structure. This discovery has founded the new, currently burgeoning field of twistrionics, offering the chance to develop new optoelectronic nanophotonic applications. In this PhD project, Moiré TMDs are studied theoretically.

In TMD bilayers, excitons, electron-hole pair quasiparticles, with high binding energies up to several 100 meV, can form due to the reduced screening and strong Coulomb interactions. This means that even at room temperature, stable long-lifetime excitons can form, and therefore the photonics of these materials is dominated by the excitons. There are various kinds of excitons which can form, be modelled and observed in TMD 2D bilayers: Dark and bright excitons, and interlayer and intralayer excitons.

This could be very versatile for new nanophotonic and spintronic applications.



The formation of excitons in 2D materials has not been fully understood so far at the fundamental level. Also, there are many open questions about the interaction of different excitons in 2D materials with each other and with other quasiparticles, such as phonons, and about their interaction with light pulses.

Our main interests in this PhD project are the formation, interaction, and light-matter interaction of trions, charged excitons composed of two electrons and one hole or two holes and one electron, and biexcitons, bound exciton-exciton pairs, in Moiré and Janus 2D TMD materials.

The PhD project can benefit from and be built on previously developed methods and experiences of the Bester group, particularly in the first half of the PhD project. Furthermore, there are many other research directions that could be intriguing and versatile.

Within the DFG SPP 2244 project, the Bester group has developed the atomic effective pseudopotential (AEP) method: a semiempirical method based on ab initio calculations to model the electronic structure of Moiré 2D TMDs. It helps to compute the relaxation of a 2D TMD Moiré bilayer and reduces the computational cost of simulating Moiré materials. Developing a configuration-interaction-based method and methodology, built on the AEP method for the relaxation, should be one focus of this PhD project.

After having developed and tested this methodology, it could be interesting to study exciton-phonon or exciton-photon interaction in these Moiré TMDs during formation, transport, and recombination of bright and dark excitons.

We'll focus particularly on Janus Moiré TMD bilayers. Janus 2D TMD materials are bilayers, composed of two different chalcogenides in each of the two single layers, which results in an internal break of the mirror symmetry within the single layers.

The results should be compared with experimental observations measured by groups at the Max-Planck-Institute of Structure and Dynamics of Matter or by external groups. The results should be published in high-ranked scientific journals.

The PhD project is integrated and will contribute new insights to the DFG priority program SPP 2244, "2D materials - Physics of van der Waals (hetero)structures.

This PhD project is expected to contribute to a better fundamental understanding of complex exciton formation and interaction in van der Waals heterostructure. This understanding can also help to develop advanced nanophotonic applications, such as excitonic LEDs or excitonic solar cells.